torch.signal.windows.gaussian#

https://pytorch.org/docs/stable/generated/torch.signal.windows.gaussian.html

Description:

Computes a window with a gaussian waveform. The gaussian window is defined as follows: The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. std (float, optional) – the standard deviation of the gaussian. It controls how narrow or wide the window is. Default: 1.0.

Function Signature

```
torch.signal.windows.gaussian(M, *, std=1.0, sym=True,
dtype=None, layout=torch.strided, device=None,
requires_grad=False)[source]#

Parameters

Keyword Arguments

Return type
```

Parameters

Keyword

std (float, optional) – the standard deviation of the gaussian. It controls how narrow or wide the window is. Default: 1.0. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric gaussian window with a standard
deviation of 1.0.
>>> torch.signal.windows.gaussian(10)
tensor([4.0065e-05, 2.1875e-03, 4.3937e-02, 3.2465e-01, 8.8250e-
01, 8.8250e-01, 3.2465e-01, 4.3937e-02, 2.1875e-03, 4.0065e-05])

>>> # Generates a periodic gaussian window and standard
deviation equal to 0.9.
>>> torch.signal.windows.gaussian(10, sym=False,std=0.9)
tensor([1.9858e-07, 5.1365e-05, 3.8659e-03, 8.4658e-02, 5.3941e-
01, 1.0000e+00, 5.3941e-01, 8.4658e-02, 3.8659e-03, 5.1365e-05])
```

Detailed Documentation

Documentation

Computes a window with a gaussian waveform.

The gaussian window is defined as follows:

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True.

M (int) – the length of the window. In other words, the number of points of the returned window.

std (float, optional) – the standard deviation of the gaussian. It controls how narrow or wide the window is. Default: 1.0.

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.
